

Mass/Spring/Damper Systems Part 1

EGR 345

GVSU

Outline

- review
- mass/spring/damper systems
 - initial condition response
 - Laplace with a second order denominator
 - Laplace with complex roots
 - critical damping
 - defining ζ (zeta)

Review: Two key characteristics of a dynamic system

- one “practical” and one mathematical
- vibrations are a transient response that can take a while to die out

Review: Mass/Spring Response

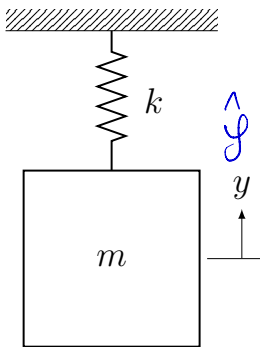
oscillate forever

- what does this system do if you give it an initial displacement and release it?
- what differential equation describes this system's dynamics?

$$y = \hat{y} + y_{\text{static}}$$

$$m\ddot{\hat{y}} + k\hat{y} = 0$$

FBD $\rightarrow$ Newton's
2nd law



$$\hat{y}(t) = \hat{y}(0) \cos(\omega t)$$

$$\omega = \sqrt{\frac{k}{m}}$$

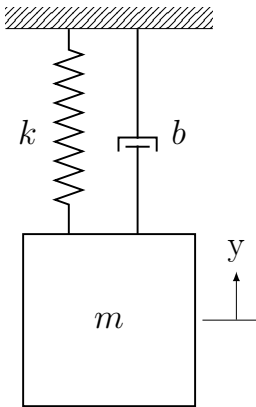
Review: What is a damper?

- what does a damper do?
- what does it represent?
- what equation models a damper?

models energy loss and
resists motion with a
force proportional to
velocity
 $f = b \dot{y}$

Mass/Spring/Damper Response

- how does this system respond to an initial displacement?



gravity is offset
by static deflection



Assume + displacement



$$m\ddot{y} = -ky - b\dot{y}$$

$$m\ddot{y} + b\dot{y} + ky = 0$$

$$y(0) = y_0$$

$$\dot{y}(0) = 0$$

$$m\ddot{y} + b\dot{y} + ky = 0$$

row 1

row 2

row 3

$$y(0) = y_0$$

$$\dot{y}(0) = 0$$

$$y(0) \quad \dot{y}(0)$$

$$\downarrow$$

$$m(s^2 Y(s) - \underline{s y_0} - \underline{0}) + b(s Y(s) - \underline{y_0}) + k Y(s) = 0$$

$$\downarrow$$

$$Y(s) = ?$$

$$\xrightarrow{\mathcal{L}^{-1}} y(t) = ?$$

$$m(s^2 Y(s) - \underline{s y_0} - 0) + b(s Y(s) - \underline{y_0}) + k Y(s) = 0$$

$$m s^2 Y(s) + b s Y(s) + k Y(s) = m s y_0 + b y_0$$

$$Y(s) (m s^2 + b s + k) = y_0 (m s + b)$$

$$Y(s) = \frac{y_0 (m s + b)}{m s^2 + b s + k} \xrightarrow{\mathcal{L}^{-1}} y(t) = ?$$

$$\frac{y_0 (ms+b)}{ms^2 + bs + k} \xrightarrow{\frac{1/m}{1/m}} \frac{y_0 (s + \frac{b}{m})}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

$$s^2 + \frac{b}{m}s + \frac{k}{m} = 0$$

from quadratic
formula

$$s = \frac{-\frac{b}{m} \pm \sqrt{\frac{b^2}{m^2} - \frac{4k}{m}}}{2}$$

define b_{cr} is when term under $\sqrt{\quad} = 0$

$$m^2 \left(\frac{b_{cr}^2}{m^2} = \frac{4k}{m} \right)$$

$$b_{cr}^2 = 4km$$

$$b_{cr} = 2\sqrt{km}$$

quadratic formula and its implications for
 $ax^2+bx+c=0$ 2nd order denominators

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

laplace

case 0:
 $b=0$, 2 purely
imaginary
roots

(mass/spring no damper)

3 cases

1. $b^2 > 4ac \rightarrow$ 2 distinct real roots

2. $b^2 = 4ac \rightarrow$ 2 repeated roots

3. $b^2 < 4ac \rightarrow$ 2 complex roots

$$y_0(s + \frac{b}{m})$$

$$s^2 + \frac{b}{m}s + \frac{k}{m}$$

$$\uparrow \quad \uparrow$$

$$2\zeta\omega_n \quad \omega_n^2$$

$$b = \zeta b_{cr}$$

$$b = \zeta 2\sqrt{km}$$

$$b_{cr} = 2\sqrt{km}$$

$$\zeta = \frac{b}{b_{cr}}$$

- $\zeta > 1 \rightarrow$ 2 distinct real roots
- overdamped

- $\zeta = 1 \rightarrow$ critical damping
- 2 repeated roots

- $0 < \zeta < 1 \rightarrow$ underdamped, complex roots

$$s^2 + \frac{b}{m}s + \frac{k}{m}$$

$$2\zeta\omega_n \quad \omega_n^2$$

$$b = 8 b_{cr}$$

$$b = 8 \cdot 2 \sqrt{km}$$

$$s^2 + \frac{2 \cdot 8 \sqrt{km}}{m} s + \omega_n^2$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\frac{s^2 + 2 \cdot 8 \sqrt{km} s + \cancel{\omega_n^2}}{\sqrt{m}^2}$$

$$s^2 + 2 \cdot 8 \sqrt{\frac{k}{m}} s + \omega_n^2$$

$$s^2 + 2 \zeta \omega_n s + \omega_n^2$$

$$Y(s) = \frac{\phi(s + \frac{b}{n})}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

if $0 < \zeta < 1$, use rows 8+9

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s+a)^2 + \omega^2$$

$$\underbrace{s^2}_{s^2} + \underbrace{2\zeta\omega_n s}_{s^1} + \underbrace{\omega_n^2}_{s^0} = \underbrace{s^2}_{s^2} + \underbrace{2as}_{s^1} + \underbrace{a^2 + \omega^2}_{s^0}$$

$$\frac{y_0(s + \frac{b}{n})}{s^2 + \frac{b}{n}s + \frac{k}{n}} = \frac{C_0 \omega}{(s+a)^2 + \omega^2} + \frac{C_1(s+a)}{(s+a)^2 + \omega^2}$$

↑

row 8

↑

row 9

$$y(t) = C_0 e^{-at} \sin(\omega t) + C_1 e^{-at} \cos(\omega t)$$